# Data bibliography / attribution

## Orbital Technosphere / Observer

### 01 CelesTrak / orbital elements and memberships

26 OMM CSV catalogs. Collected 2026-11-03T17:13:12Z to 2026-11-03T17:14:05Z.
Active catalog supplies the satellite population; membership catalogs identify named fleets. Three bounded debris catalogs supplement it.
Catalogs: active, starlink, oneweb, kuiper, qianfan, hulianwang, gnss, iridium-next, globalstar, orbcomm, resource, science, stations, fengyun-1c-debris, iridium-33-debris, cosmos-2251-debris, planet, spire, intelsat, ses, eutelsat, telesat, gps-ops, glo-ops, galileo, beidou.
Attribution: CelesTrak. Exact feed URLs, element epochs and hashes are retained in the source ledger.
https://celestrak.org/NORAD/documentation/gp-data-formats.php

### 02 SGP4 / Vallado, Crawford, Hujsak and Kelso

Positions propagated from the pinned elements using the reference SGP4 implementation, WGS-72 and AFSPC mode.
Calculation instant: 2026-11-03T17:10:00Z. 44 records excluded by epoch; 0 propagation failures.
Public element uncertainty and approximate Earth rotation limit these visualization positions.
https://celestrak.org/publications/AIAA/2006-6753/

### Historical validation / NASA SSCWeb

Reference instant: 2026-08-13T15:36:00Z.
Historical regression reference only. The November refresh was unreachable; this response does not provide an independent comparison of the plotted November positions.
https://sscweb.gsfc.nasa.gov/WebServices/REST/

### Cartography / Cartofreako

Projection: Authagraph. The projection bibliography credits its mathematical and design sources.
Separate map and legend composition follows the Setouchi ST2 layout. Typography: Atkinson Hyperlegible; bundled fonts under the SIL Open Font License.
https://bdekoz.github.io/cartofreako/docs/pages/projections/authagraph/bibliography.html

### Pinned inputs / source ledger

The profile records the calculation time, observer, filters and every catalog URL. Adjacent acquisition.json and SHA256SUMS record retrievals and original input bytes.
Map and legend carry the same timestamp and counts; the complete acquisition ledger is embedded in this SVG.
Map image scaled to 98.863636% with proportional sheet margins.
https://github.com/bdekoz/cartofreako/blob/main/assets.static/orbital-technosphere/20261103/orbital-technosphere-profile.json

Setouchi AuthaGraph: +4.35-inch printed horizontal move on a 43.5-inch image. Celestial directions retain source registration in the same cyclic frame.

### MSP speculation 4 / Midnight blue / champagne

Randoma11y: ground #010434; marks/type #ead784. sRGB contrast 13.62:1 (screen only).
https://randoma11y.com/%23ead784/%23010434
Epson Stylus Pro 9000 / metallic glossy paper / cyan and light magenta reported low; grays reportedly print violet.
Intended RGB hues; final printed hue is unmeasured. No green correction, ink separation or metallic reflection simulation.

## SGP4 positions / 2026-11-03T17:10:00Z

1,088 plotted objects / named fleets and residual roles

*Seven-day element age; six-hour future tolerance.*

Starlink 549
OneWeb 44
Kuiper 31
Qianfan 20
Hulianwang 21
Iridium 5
Globalstar 4
ORBCOMM 0
Planet 0
Spire 2
Intelsat 38
SES 26
Eutelsat 7
Telesat 12
GPS 11
GLONASS 8
Galileo 13
BeiDou 13
Debris 53
Other active 220
Other navigation 2
Other Earth observation 6
Science 2
Human presence 1

One object per NORAD ID. Overlapping named memberships use profile order.

Name and locpoint sublayers are independent. Names are hidden by default.

Black x: debris. Sunburst: Starlink. Square: Intelsat. Triangle: remaining Earth observation.

Active-marker opacity records illumination. Debris remains solid black.

Topocentric celestial directions above the San Francisco horizon. Equator and horizon are reference lines.

Starlink sunbursts: 1.5× linear size on both orbital maps.

# Orbital technosphere Observer / Authagraph